Photocatalysis Degradation of Organic Pollutants from Water with a Nanomaterial Heterojunction

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Introduction

PFAS

- The U.S. Environmental Protection Agency reports there are about 15,000 types of PFAS compounds, known as forever chemicals.
- These can be found in dental floss, nail polish, cleaning products, and more (Fig. 1; Míšľanová & Valachovičová, 2025).
- PFOA is one of the most common long-chain PFAS (Wei et al., 2025).

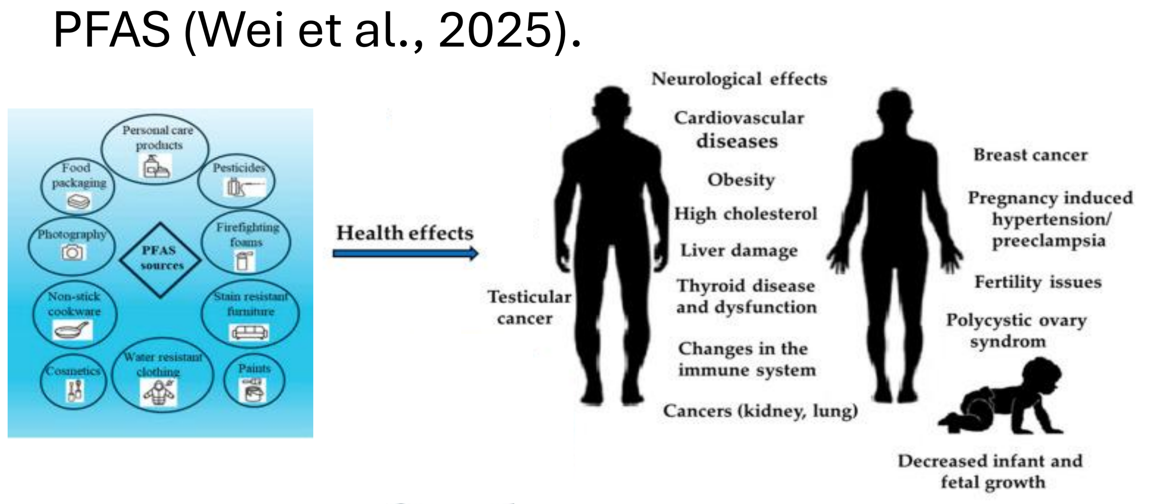

Figure 1: PFAS sources and health effects. Modified from Míšľanová & Valachovičová, 2025.

Photocatalysis

- A method to degrade PFAS is through photocatalysis.
- As light is absorbed by the semiconducting photocatalyst, an electron-hole separation occurs.
- These charges generate reactive oxygen species, which can oxidize PFAS.

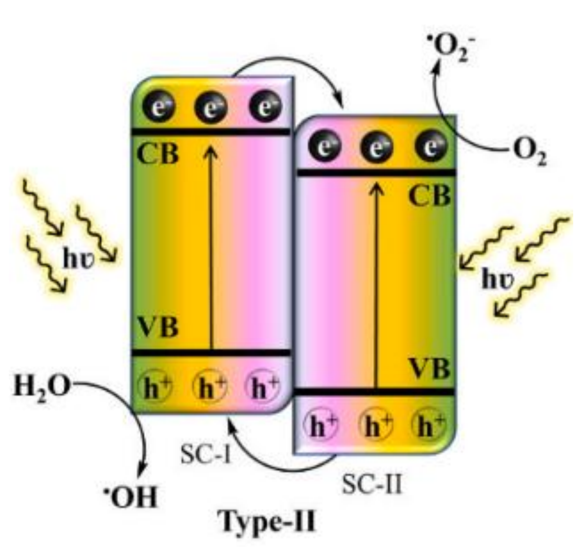

Figure 2: Type-II heterojunction proposed for our materials (Tiwari et al., 2025).

- Electron-hole separation life-time can be elongated with a heterojunction (Fig. 2).

Electromigration

- Photocatalysis is greatly dependent on the structure of the materials.
- Understanding the mechanisms of synthesis is important as this is what determines the structure of the materials.
- Electromigration (EM) one of the governing mechanisms in our synthesis method. EM is the displacement of atoms due to momentum transfer from electrons moving through a material (Fig. 3; Fernández, 2024).

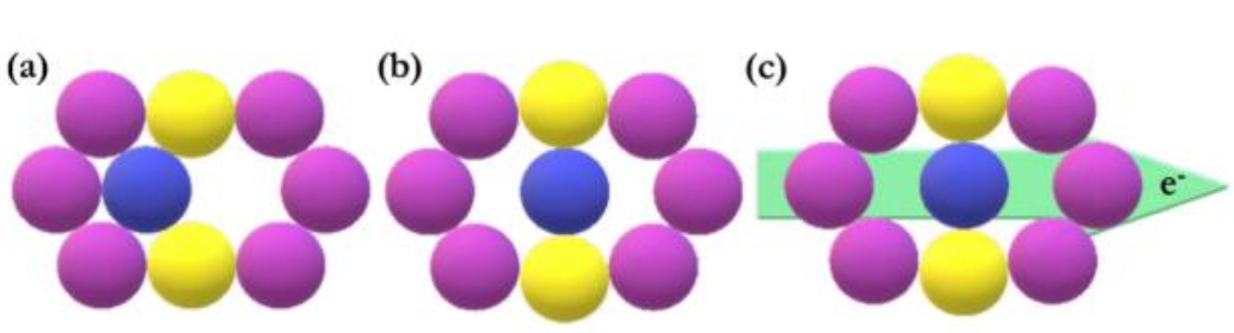

Figure 3: (a) atom in stable position, (b) atom in activated position by random processes, and (c) atom in activated position by electromigration (Fernández, 2024).

Methods

Synthesis

- The Joule heating synthesis setup had the following parameters (Fig. 4):

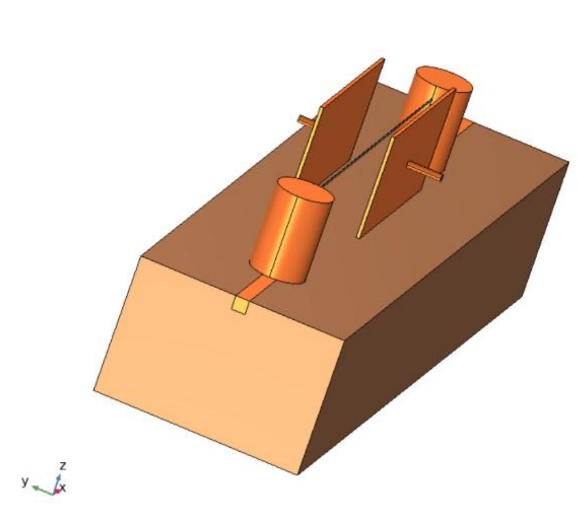

Figure 4: Synthesis apparatus.

- 4.5-5.5 A through Mo wire.
- ~5.0 mm Cu plate separation.
- 200 V applied to Cu plates.
- ~6.0 cm Mo wire length.
- Time ~15 secs.
- 5 hrs. 250 rpm in mill.

- When EM occurs, Mo ions in the wire migrate to an activated position where they are more likely to bond with the room oxygen atmosphere.
- The molybdenum oxide species from this reaction then volatilize and get deposits on the Cu electrodes.
- It has been shown that this EM process is driven by electric current density and thermal gradients (Fernández, 2024).

Simulation

Conservation of Current in the wire is imposed with the continuity equation

$$-\nabla \cdot (\sigma \nabla V) = 0,$$

where σ is the electrical conductivity of the material and V is the electric potential.

The Joule heating calculated with

$$Q = \sigma |\nabla V|^2$$

is used as a source in the heat transfer equation

$$\rho\, C_p \left(\frac{\partial T}{\partial t} + \vec{u} \cdot \nabla T \right) - \varkappa \nabla^2 T = Q,$$

where ρ is the density, C_p is the specific heat capacity, T is the temperature, $\vec{u}$ is the velocity field of the air, and $\varkappa$ is the material's thermal conductivity.

From the heat transfer equation, the temperature T can be found and finally the thermal deformation can be calculated with

$$\varepsilon = \alpha (T - T_{ref}),$$

where ε is strain, α is the thermal deformation constant, and T_{ref} is the initial temperature of the wire.

Results

Characterization

- Characterization was done on molybdenum trioxide (MoO_3) and titanium dioxide (TiO_2) after ball milling.
- The scanning electron microscope (SEM) image shows the laminar and spherical structures associated with the α and β phases of MoO_3, respectively, and spherical structures for anatase TiO_2 phase (Figs. 6a & 6b).
- Micro-Raman spectroscopy of MoO_3 showed the vibrational modes associated with the α and β phases, as well as the m-MoO_2 conducting phase from defects in the material. TiO_2 was also measured to ensure no phase transition occurred; all the vibrational modes were associated with the anatase phase (Figs. 6c & 6d).
- X-ray diffraction (XRD) measurements revelated the atomic planes associated with the α and β phases for MoO_3, and average particle size ~40 nm for both phases (Fig. 6). For TiO_2 the anatase atomic planes were found and the average particle size to be ~60 nm.

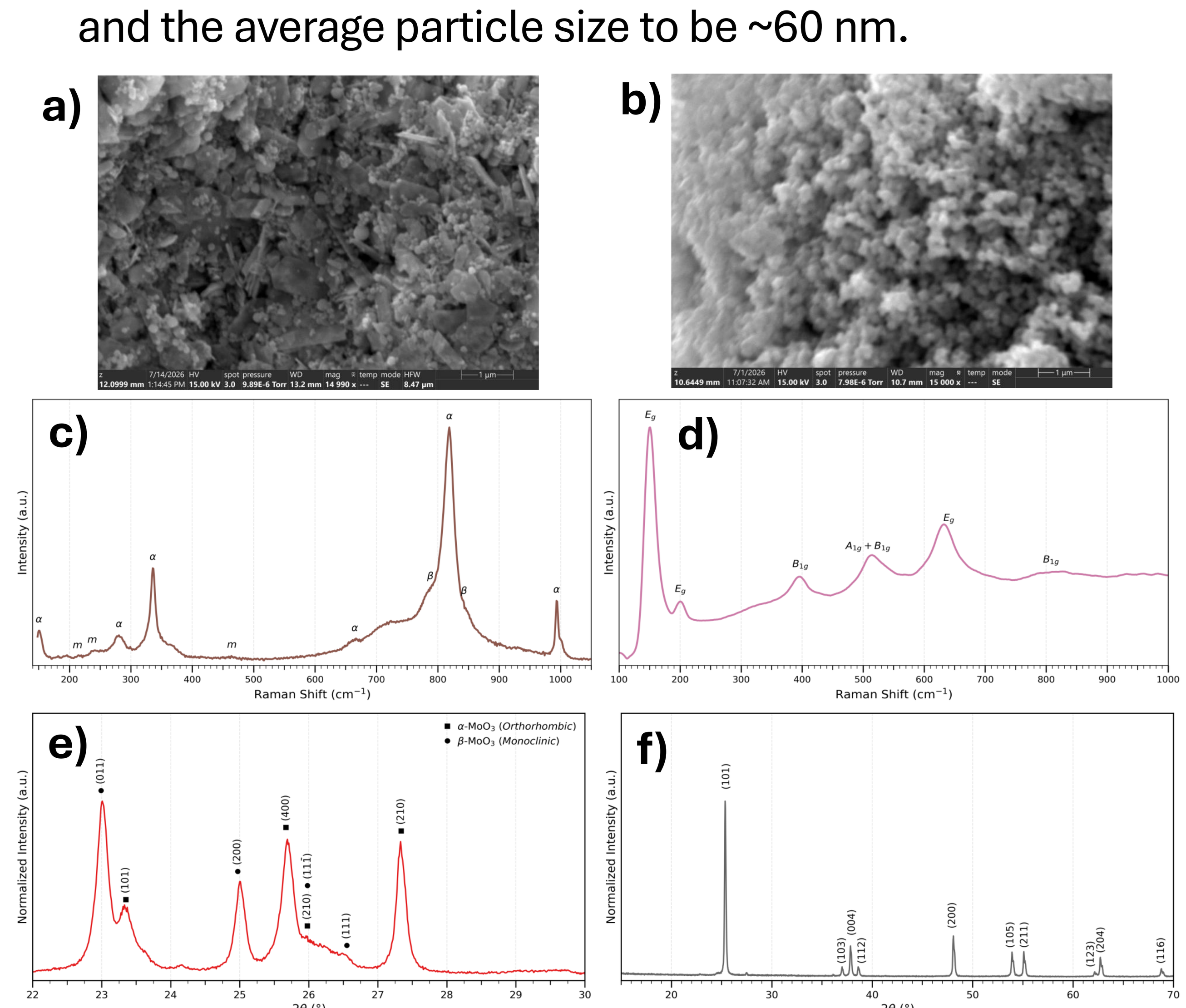

Figure 6: (a) & (b) SEM images of MoO_3 and TiO_2, respectively. (c) & (d) micro-Raman spectra for MoO_3 and TiO_3, respectively. (e) & (f) XRD data for MoO3 and TiO_2 respectively.

Simulation

- The COMSOL simulation gave insights into the EM mechanisms, allowing us to calculate the average unit cell sizes from the deformation data (Fig. 7).

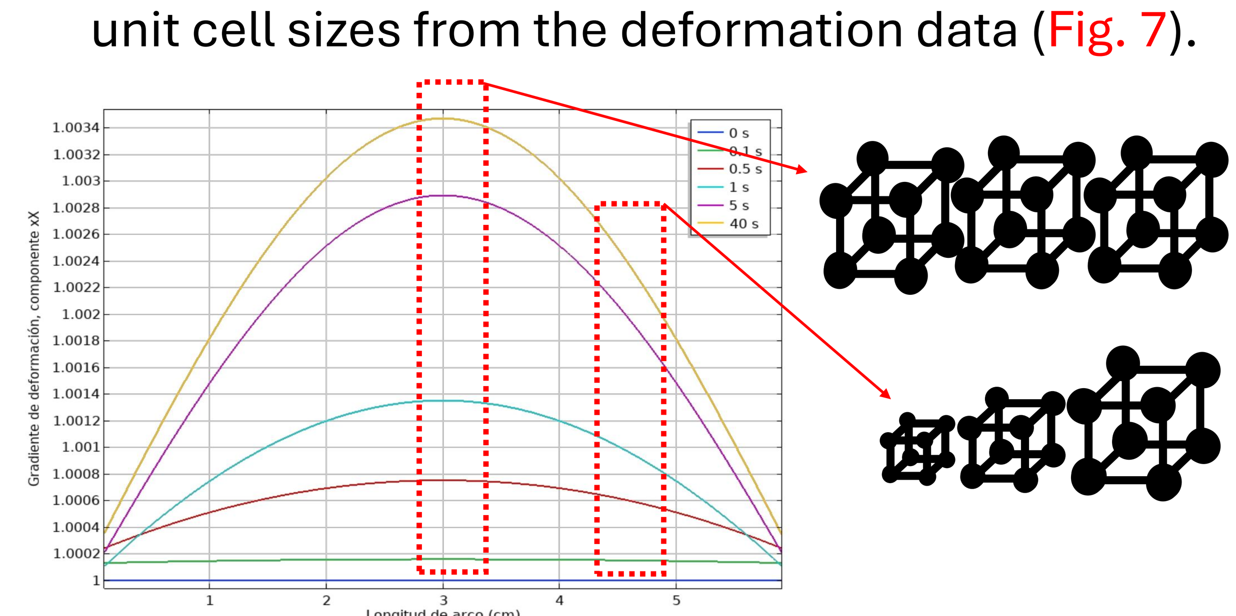

Figure 7: XX component of the deformation gradient in the y-axis and length of wire in the x-axis. The different functions plotted correspond to different simulation times in seconds.

Future Direction

Density Functional Theory

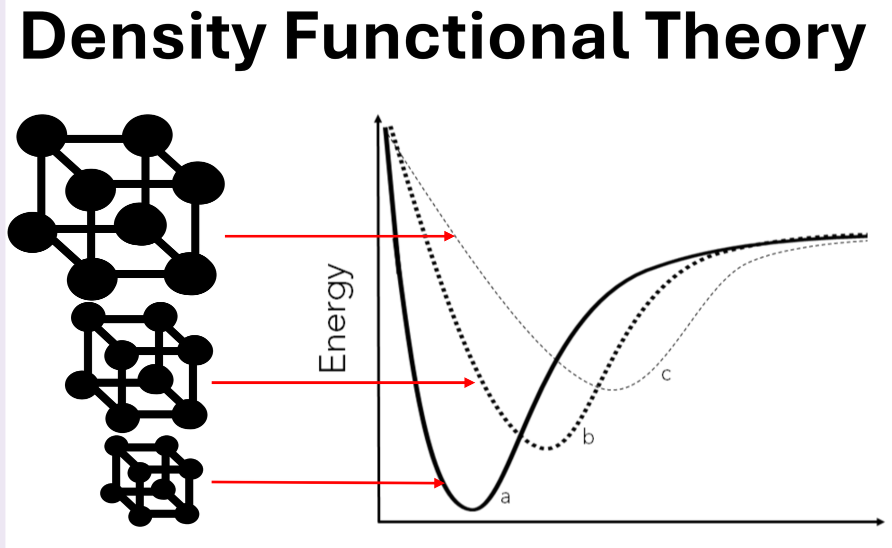

Figure 8: Hypothesis of unit cell size effect in energy needed to result in EM.

- Utilize the unit cell sizes as entries for Density Functional Theory (DFT) calculations.
- Investigate the relationship between potential energy curves and unit cell size (Fig. 8).

Creating the Heterojunction

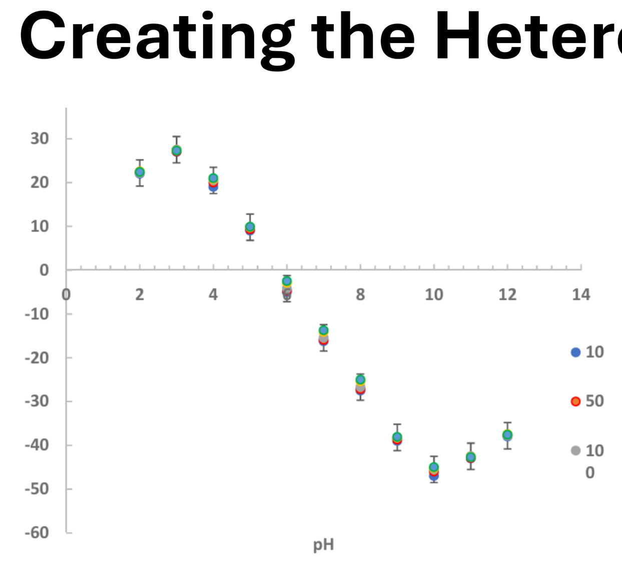

Figure 9: Results for different values of TiO_2 concentrations (mg/L) (Marsalek et. al., 2021).

- Find the pH at which TiO_2 and MoO_3 have opposite polarities (Fig. 9).
- Then, use this pH to have the heterojunction form by electrostatic self-assembly.

Monitoring PFOA degradation

- Quantify photocatalysis degradation (Fig. 10).

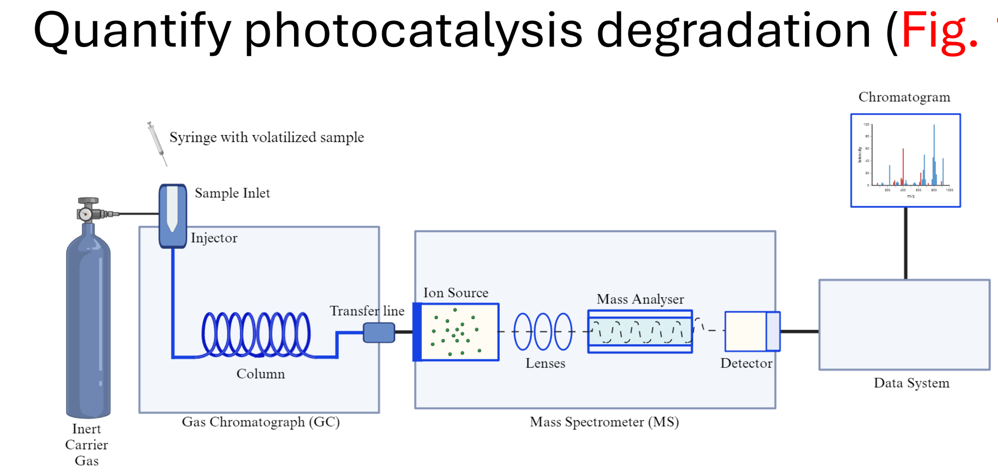

Figure 10: Proposed strategy, gas chromatography mass spectrometry setup, for monitoring PFOA degradation.

Acknowledgements

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